

Quantum Random Walk

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Abstract—Quantum walk can be viewed as the adaptation of Markov chain in the quantum world. In this paper, we define the discrete time quantum walk. We show some differences with the classical random walk. Then we examine a search algorithm which uses quantum walk, before discussing the potential use of quantum walk in quantitative finance.

I. INTRODUCTION

When the extraordinary properties of the quantum world were discovered, like superposition of states and entanglement, people tried to use it in computer science. This led to the exceptional result of the Shor's algorithm, that can factorize integers faster than any algorithm on a classical computer [1]. This gave to the quantum computer science a great renown. As Markov chain is a powerful tool in many fields, people defined an analogous instrument in the quantum world, called the quantum random walk. Likewise Markov chains, there are continuous time and discrete time quantum walks. In this paper, we shall only consider discrete time quantum walk. After introducing the definitions, we examine an application that works faster than its classical counterpart. This algorithm finds an element in a list with a particular property. Finally, we examine a possible application of quantum walk in finance.

II. DEFINITION

Like Markov chains, Quantum random walks have a graph $G = (\mathcal{V}, \mathcal{E})$ as support, where $\mathcal{V}$ is the set of the vertices and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ the set of the edges. We denote $N = |\mathcal{V}|$ the number of vertices, with $1 \leq N \leq +\infty$. The aim is to find a definition analogous to Markov chain. Let us consider for example an infinite line. We can take $\mathcal{V} = \mathbb{Z}$ and $\forall (x, y) \in \mathbb{Z}^2, (x, y) \in \mathcal{E} \Leftrightarrow |x - y| = 1$. With Markov chain, we start at the position 0, and at each iteration, we move left with probability $p \in [0, 1]$ or right with probability $1 - p$. Let us move on to the quantum world.

Definition 1. A state is a unitary vector of an Hilbert space $\mathcal{H}$. If there are N possible states, or in our case N vertices, we shall choose a separable Hilbert space of dimension N , denoted by $\mathcal{H}_N$. We denote by $|n\rangle$ the unitary vector of which every component is zero but the n -th one.

We choose the initial state to be $|0\rangle$. At each iteration, the state will be a combinaison of the nearest vertices. The first idea would be to go from $|n\rangle$ to $a|n-1\rangle + b|n\rangle + c|n+1\rangle$ at each iteration. The coefficients $|a|^2, |b|^2$ and $|c|^2$ are respectively the amplitude of going left, staying in the same state and going right. However, this is not possible [2].

Theorem 1. An unitary transformation verifies the previous equation if and only if one of these conditions is true:

- $|a|=1, |b|=|c|=0$
- $|b|=1, |a|=|c|=0$
- $|c|=1, |a|=|b|=0$

Therefore, as all transformations in the quantum world are reversible and unitary, we cannot choose such a definition. The common way of solving this problem is to introduce a coin state. To simplify, we shall now only consider graphs that are d -regular. We now consider two Hilbert spaces: the position espace $\mathcal{H}_N$ and the coin space $\mathcal{H}_d$. The state of the system will be a combinaison of the state in $\mathcal{H}_N$ and $\mathcal{H}_d$. Thus, we shall denote $\mathcal{H} = \mathcal{H}_N \otimes \mathcal{H}_d$ the Hilbert space of the states.

Let us go back to the case of the infinite line. At each iteration, we apply two unitary transformations: C and S . The operator C changes the state of the coin, and S applies a shift corresponding to the coin state.

Definition 2. We denote $|L\rangle$ the going-left state and $|right\rangle$ the other one. Instead of writing $|n\rangle \otimes |L\rangle$, we shall prefer the shorter expression $|n, L\rangle$. We define C so that $C|n, L\rangle = a|n, L\rangle + b|n, R\rangle$ and $C|n, R\rangle = c|n, L\rangle + d|n, R\rangle$. The operator S does the basic operations $S|n, L\rangle = |n-1, L\rangle$ and $S|n, R\rangle = |n+1, R\rangle$.

One can remark that C does not affect the position state, and S does not change the coin state. Besides, C can simply be written $Id \otimes C_0$ where $C_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

III. QUICK COMPARISON BETWEEN CLASSICAL AND QUANTUM WALKS ON AN INFINITE LINE

We keep the definitions of the previous section. For the classical walk, we take $p = 1/2$. For the quantum walk, we choose the Hadamard coin $C_0 = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$.

To begin with, let us see what happens with the initial state $\frac{1}{\sqrt{2}}|0, L\rangle + \frac{1}{\sqrt{2}}|0, R\rangle$. This seems to be justified as we want the walk to be unbiased. After the application of C we obtain $|0, L\rangle$ and thus after S we obtain $|-1, L\rangle$. The result is one step to the left, and a change of the coin state. To have a better comparison between the two walks, we thus prefer to use $|0, L\rangle$ as initial state. Let us see what happens after each operation, keep in mind that two operations count as one step of the classical walk.

- $|0, L\rangle$
- $\frac{1}{\sqrt{2}}|0, L\rangle + \frac{1}{\sqrt{2}}|0, R\rangle$
- $\frac{1}{\sqrt{2}}|-1, L\rangle + \frac{1}{\sqrt{2}}|1, R\rangle$
- $\frac{1}{2}|-1, L\rangle + \frac{1}{2}|-1, R\rangle + \frac{1}{2}|1, L\rangle - \frac{1}{2}|1, R\rangle$
- $\frac{1}{2}|-2, L\rangle + \frac{1}{2}|0, R\rangle + \frac{1}{2}|0, L\rangle - \frac{1}{2}|2, R\rangle$
- $\frac{1}{2\sqrt{2}}(|-2, L\rangle + |-2, R\rangle + 2|0, L\rangle - |2, L\rangle + |2, R\rangle)$
- $\frac{1}{2\sqrt{2}}(|-3, L\rangle + |-1, R\rangle + 2|-1, L\rangle - |1, L\rangle + |3, R\rangle)$

At the third and fifth lines, the results are identical to those of the classical walk. However, at the seventh line, the quantum interferences lead to a result slightly different from the classical one. In fact, if we were to pursue the iterations, we would eventually remark that a quantum walk can spread quadratically faster than the classical walk, depending on the choice of the initial states and the coin [3].

IV. A QUANTUM SEARCH ALGORITHM

A. Modelling of the problem

In this section we will describe an alternative to the Grover's algorithm based on quantum walk. The goal of these algorithms is to find the unique input in a list of size N that has a given image by a function. For instance, let us imagine that we want to find in a directory the name of someone whom we know the phone number. As the list is unordered, the classical is in $O(N)$. However, in the quantum world there exists an algorithm discovered by Grover that only takes $O(\sqrt{N})$ evaluations of the function [4].

The algorithm we present in this section is largely inspired by the Grover's algorithm, but the fact that it uses quantum walk provides some interesting differences. It was discovered by Shenvi, Kempe and Whaley [5].

Definition 3. We define $\mathcal{G}$ as the hypercube with $N = 2^d$ vertices. Each vertex can be expressed as a d -bit binary string. We define the set of the edges such that $\forall (x, y) \in \mathcal{V}^2, (x, y) \in \mathcal{E} \Leftrightarrow |x - y| = 1$ where $|\cdot|$ is the Hamming weight. Thus, $\mathcal{G}$ is n -regular. The Hilbert space considered is $\mathcal{H}_{2^d} \otimes \mathcal{H}_d$.

Definition 4. We denote by $(e_i)_{1 \leq i \leq d}$ the basis vectors of the hypercube. The operator S is simply the adaption of the previous one but with n possible directions: $\forall 1 \leq i \leq d, S|n, i\rangle = |n + e_i, i\rangle$.

One can remark that S verifies the equation $S = \sum_{i=1}^d \sum_n |i, n + e_i\rangle \langle i, n|$. For the coin matrix, we use one that is called the "Grover's diffusion operator": $C_0 = -Id + 2|S_C\rangle \langle S_C|$ where $S_C = \frac{1}{\sqrt{d}} \sum_{i=1}^d |i\rangle$. Thus, C_0 corresponds

to the equal superposition over all directions. We define $U = SC$ the unitary evolution operator. We denote $|\psi_0\rangle$ the equal superposition over all states, that is $|\psi_0\rangle = |S_P\rangle \otimes |S_C\rangle$ where $|S_P\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N |n\rangle$. Thanks to the special form of C , one can calculate the eigenvalues and eigenvectors of U to show that $|\psi_0\rangle$ is an eigenvector of U with eigenvalue 1. Therefore, the operator U leave the state $|\psi_0\rangle$ unchanged. The idea of the algorithm is to put the system in the state $|\psi_0\rangle$, then apply an operator U' quite similar to U , that will mainly affect the targeted state. Without loss of generality we shall consider that the targeted vector is $0_{\mathcal{H}_N}$. We then take $C' = Id \otimes C_0 + |0\rangle \langle 0| \otimes (C_1 - C_0)$ as a coin. We choose $C_1 = -Id$ by analogy with the Grover's algorithm. The choice of C_1 will be more particularly described later. We can detail the operator $U' = SC' = S(Id \otimes C - |0\rangle \langle 0| \otimes (C - Id)) = U - 2S(|0\rangle \langle 0| \otimes |S_C\rangle \langle S_C|)$.

B. Description of the algorithm

We have to find the unique input that verifies a given equation amongst $N = 2^d$ possible choices. We associate each of the possible choices with one of the vertices of the hypercube $\mathcal{G}$. To simplify the writing, we suppose that the targeted vertex is $0_{\mathcal{H}_N}$. One of the biggest advantage of the quantum world is the fact that we have at our disposition the coin C' , called the *oracle*, that can act on the targeted state without making a measurement, that would destroy the quantum superposition. The algorithm works as follow:

- 1) Initialise the quantum computer to the equal superposition of states $|\psi_0\rangle = |S_P\rangle \otimes |S_C\rangle$. This can be done by applying N times the single-bit Hadamard operation to the $|0_{\mathcal{H}_N}\rangle$ state, and d times to the $|0_{\mathcal{H}_d}\rangle$ state. The Hadamard operator is the extension for any dimension of the Hadamard coin presented in the previous section.
- 2) Apply t times the unitary operator $U' = SC'$, where t is the nearest integer of $\frac{\pi}{2}\sqrt{N}$.
- 3) Make a measurement in the $|n, i\rangle$ basis.

Shenvi, Kempe and Whaley showed that the algorithm gives the good solution with probability $\frac{1}{2} - O(\frac{1}{d})$. Thus one can find the marked state with a probability of error as small as wanted by repeating the algorithm. The proof is not straightforward, and is too long for this article. The idea of the proof is to switch the framework from the hypercube to an equivalent infinite line, then to operate on the eigenvalues of U' to demonstrate some properties, before concluding.

C. Comparison with the Grover's algorithm

There are some strong similarities between the two algorithms. Both begin by the initialisation in the equal superposition of all states. Both algorithms use an oracle that marks the targeted state with a phase of -1 . Both algorithms are in $O(\sqrt{N})$, and we have to make the measure at a particular time to obtain the highest probability of finding the marked state. However there are some fundamental differences. In the Grover's algorithm, all the non-marked states have an equal probability to be found. In the quantum walk algorithm, the nearest vertices of the targeted one have higher probability

to be found. Thus this algorithm depends on the underlying topology of the graph. One improvement is the fact that the oracle acts on a d -dimensional Hilbert space, instead of the 2^d -dimensional searching space of the Grover's algorithm. This could be very useful if a physical implementation were to exist. An other difference is the choice of the oracle. With the quantum walk algorithm, we chose $C_1 = -Id$, giving the same change of phase as in the Grover's algorithm. However, there are no analytical result for other coins, and there might be some reductions of the complexity, within a constant factor.

V. A POSSIBLE APPLICATION IN FINANCE

A. Definitions

All the definitions and concepts used in this section are explained with precise details in the book *Options, Futures, and Other Derivatives* by John C. Hull, that is considered as one of the best references in the financial domain [6]. It is a harsh but potentially very profitable problem to know what will be the price of an asset or a financial derivative in the future. One could buy assets when the price will rise, or short them when it is going to fall. Some models have been developed on many different financial products. We shall only examine the more famous one, the *Black-Scholes model*, for nearly all models are inspired by it. In spite of its hegemonic position, all agree that the Black-Scholes model suffers from intrusive conditions. Therefore, after having explained the model, we shall imagine an improvement, based on quantum walk. This model allow us to evaluate the price of an European-style option.

Definition 5. *The buyer of an European-style option have the right (but is not obliged) to buy or sold a predefined quantity of a certain stock, at a predetermined stock price K , at the end of a time lapse T . It is different of the American-style option which can be exercised at any time before the expiration date. If the buyer can buy a quantity of a stock, it is called a call option. If it is for selling, it is called a put option.*

Let us take an illustrative example. There are two companies, A and B . A have to buy 100 units of a stock X in a year. At the moment, the price of X is 10\$ per unit. Let us suppose that A believes that the stock price will fall, but cannot afford to be wrong. Furthermore, let us suppose that B strongly assume that the price will rise. In this case, A can buy to B an option for 100 units at 10\$ per unit, for a price that we suppose to be 20\$. If in a year the stock price has fallen to 8\$, A will not exercise the option, and normally buy the 100 units for 800\$. It is a total cost of 820\$ for A , instead of the 1000\$ it should have paid now, not counting the storage cost. However, if the stock price has risen to 15\$, A will exercise the option, and buy 100 units for 1000\$. It would be a total cost of 1020\$, that is more than just having bought it directly, but is far less than the 1500\$ it should have paid in a year. Thus, options can be used to hedge the risks. It is very difficult to decide the price of an option, that we chose to be 20\$ in our example. That is the aim of the Black-Scholes model.

Definition 6. *The arbitrage is the possibility for traders to make a riskless profit. This is obtained by the differences that can exist between two markets. Buy buying in one market and selling in another one, it is sometimes possible to benefit from the situation. People who use this to make a living are called arbitrageurs.*

B. assumptions of the Black-Scholes model

The assumptions of the model on the assets are:

- The rate of return on the riskless asset is constant. It is called *risk-free interest rate* and is denoted by r . The risk-free interest rate is not an easy concept and will not be detailed here.
- The instantaneous log return of stock price is a geometric Brownian motion. The drift and volatility are supposed to be constant. We denote by S the stock price and by σ the volatility of the log return of the stock price.
- The stock does not pay a dividend.

There are also some assumptions made on the market:

- There are no possible arbitrage actions. We consider that the arbitrageurs are efficient enough to make immediately disappear any arbitrage possibility.
- It is possible to borrow, to lend, to buy and to sell any fractional amount of cash.
- There are no fees for such transactions.

All of these assumptions are questionable, but before analysing the one that interests us most, we present the result for the curiosity of the reader. We denote by $C(S, t)$ the price of an European call option. We have the partial differential equation $\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS \frac{\partial C}{\partial S} - rC = 0$. By integrating and considering the boundary conditions, we obtain $C(S, t) = N(d_1)S - N(d_2)Ke^{-r(T-t)}$ where N is the standard normal cumulative distribution function, and d_1, d_2 are some funtions of parameters S and the current time t . The expressions of d_1, d_2 is not easy to read and thus omitted, but it is possible to easily compute the price of a call option. The formula for a put option is quite similar.

C. Main criticism

Amongst all the assumptions made, the most criticised is the modelling of the log return of the stock price. As seen in Figure 1, during crisis and krachs, the stock prices are not following a normal law, not even a lognormal law. The volatility can become very high, and the autocorrelation very strong.

In 2008, when the famous bank Lehman Brothers was filing for bankruptcy, some risk managers at WestLB bank had by accident the idea of modelling the stock price of an asset by a quantum walk [7]. One of the trainees had unfortunately replaced $t = \sqrt{252}$ by $t = 252$ when calculatating the annual volatility (there are in average 252 "trading days" per year). He had a volatility in $O(t)$ instead of the common $O(\sqrt{t})$ when dealing with Wiener processes. It is somewhat reminiscent of the conclusion of Section III. Indeed, a Wiener process is the scaling limit of a random walk on a line. The

risk managers, seeing that the simulations were close to the graphics of Lehman Brothers stock price, asked themselves three questions:

- 1) What if the volatility of an asset did not scale with the square root of time but with time itself?
- 2) What if the random walk did not converge to any limiting distributions?
- 3) What, if via some tunneling mechanism an asset reaches a state, say, an absorbing state, at an exponential speed?



Figure 1. The stock price of Lehman Brothers before it collapsed, showing that a stock price cannot be modelised by a geometric Brownian motion.

Without knowing it, they exactly depicted the properties of quantum walk. Moreover, thanks to quantum interferences, the walk may spread slower or faster at some times. In the research field of quantum finance, some things have already been tried, but for now nobody really investigated into the possibility of using quantum walk. With the good underlying graph, researchers may provide a model that predicts the many crisis and krachs, or might be consistent with the autocorrelation that one can observe on the graphics of stock prices.

VI. CONCLUSION

We defined the discrete time quantum walk, that is an analogous to Markov chain. We showed some the differences between classical and quantum walks on an infinite line, namely the diffusion spread and the biased evolution. Then we saw an application of the quantum walk. By using it, one could find quadratically faster a certain element in a list than with classical algorithms. Finally, we presented a possible application in finance and its motivation.

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